

Solving for L_M Step-by-Step

Specific-Factors Model

Setup

Home produces Manufactures (M) and Food (F). Labor is mobile between sectors, while capital is specific to M and land to F .

$$Q_M = K^{1/2}L_M^{1/2}, \quad Q_F = T^{1/2}L_F^{1/2}, \quad L_M + L_F = L.$$

Endowments:

$$K = 100, \quad T = 64, \quad L = 100.$$

Prices: initially $P_M = 2$, $P_F = 1$. We will later examine the case when $P_M = 3$.

The equilibrium condition (labor mobility):

$$P_M \cdot MPL_M = P_F \cdot MPL_F,$$

where

$$MPL_M = \frac{\partial Q_M}{\partial L_M} = \frac{1}{2}K^{1/2}L_M^{-1/2}, \quad MPL_F = \frac{\partial Q_F}{\partial L_F} = \frac{1}{2}T^{1/2}L_F^{-1/2}.$$

Since $K^{1/2} = 10$ and $T^{1/2} = 8$:

$$MPL_M = 5L_M^{-1/2}, \quad MPL_F = 4L_F^{-1/2}.$$

Also $L_F = 100 - L_M$.

Case 1: $P_M = 2$, $P_F = 1$

Step 1: Wage equalization condition

$$2 \cdot 5L_M^{-1/2} = 1 \cdot 4(100 - L_M)^{-1/2}.$$

Simplify:

$$10L_M^{-1/2} = 4(100 - L_M)^{-1/2}.$$

Step 2: Square both sides to remove the square roots

$$100L_M^{-1} = 16(100 - L_M)^{-1}.$$

Step 3: Cross-multiply

$$100(100 - L_M) = 16L_M.$$

Step 4: Expand and solve for L_M

$$10,000 - 100L_M = 16L_M \quad \Rightarrow \quad 10,000 = 116L_M.$$

$$\boxed{L_M = 86.21, \quad L_F = 13.79.}$$

Step 5: Compute the common wage

$$w = P_M \cdot MPL_M = 2 \cdot 5L_M^{-1/2} = 10L_M^{-1/2}.$$

$$w = 10/\sqrt{86.21} \approx 10/9.29 \approx \boxed{1.08.}$$

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Case 2: $P_M = 3, P_F = 1$

Step 1: Wage equalization

$$3 \cdot 5L_M^{-1/2} = 1 \cdot 4(100 - L_M)^{-1/2},$$

$$15L_M^{-1/2} = 4(100 - L_M)^{-1/2}.$$

Step 2: Square both sides

$$225L_M^{-1} = 16(100 - L_M)^{-1}.$$

Step 3: Cross-multiply

$$225(100 - L_M) = 16L_M.$$

$$22,500 - 225L_M = 16L_M.$$

Step 4: Solve for L_M

$$22,500 = 241L_M \quad \Rightarrow \quad \boxed{L_M = 93.36, \quad L_F = 6.64.}$$

Step 5: Compute new wage

$$w = P_M \cdot MPL_M = 3 \cdot 5L_M^{-1/2} = 15L_M^{-1/2}.$$

$$w = 15/\sqrt{93.36} = 15/9.66 \approx \boxed{1.55.}$$

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Interpretation

- As P_M rises, L_M increases (from 86.2 to 93.4), meaning more labor moves to manufacturing.
- The wage rises (from 1.08 to 1.55) but less than the increase in P_M (which rose 50%).
- Capital (specific to M) gains the most because it benefits from both the price increase and more labor.
- Land (specific to F) loses because L_F falls and P_F is unchanged.
- Workers have an ambiguous outcome: their real wage in terms of F rises, but in terms of M falls.